Abdul Kareem

| abdulkareemakn@proton.me | LinkedIn: Abdul Kareem | GitHub: abdulkareemakn

Education

COMSATS University Islamabad — Lahore, Pakistan
Bachelor of Science in Software Engineering — Sep. 2024 - Sep 2028

Punjab Group of Colleges — Lahore, Pakistan
Intermediate in Computer Science — Aug. 2022 - Aug 2024

Projects

Ayak | TanStack Start, Cloudflare Workers, D1, R2 — 2026 - Present

- Developed a full-stack e-commerce platform for a homemade jewelry store using TanStack Start, deployed on Cloudflare Workers
- Designed a relational schema with Cloudflare D1 (SQLite) for products, orders, and payments, with Zod for end-to-end schema validation
- Implemented bank-transfer payment verification workflow with proof-of-payment uploads via Cloudflare R2, and an authenticated checkout flow using email OTP-based authentication

Cursus | TanStack Start, React, Drizzle — 2026 - Present

- Built a full-stack web application that auto syncs university timetables to students' calendars
- Designed and implemented a custom PDF-parsing pipeline (Python, pdfplumber) extracting structured schedule data to be used programmatically
- Designed a data ingestion pipeline via CI/CD to automatically fetch the timetable, parse it, and update ICS calendar feeds automatically

exRES | Python, scikit-learn, XGBoost, pandas — 2025

- Built an ML pipeline predicting English Premier League match outcomes across 8,270 historical matches, benchmarking Logistic Regression, Random Forest, and XGBoost under two feature-set configurations
- Designed a strict temporal train/validation split (2002/03 - 2023/24 training, 2024/25 held out) using scikit-learn Pipelines to prevent preprocessing leakage
- Achieved 53.4% match-outcome accuracy and a 0.85 Spearman rank correlation between predicted and actual final standings, benchmarked against a bookmaker-odds baseline
- Built a static HTML/CSS/JS dashboard to visualize predicted vs. actual league standings from model output

Experience

Self-Hosted Infrastructure & Homelab — July 2025 - Present
Personal Project — Lahore, Pakistan

- Deployed and maintained various containerized services (Navidrome, Memos, OpenGist, Litestream) on a DigitalOcean VPS using Docker Compose
- Configured Caddy as a reverse proxy with Cloudflare DNS and TLS termination, securing routes via Auth0 OIDC and Tinyauth, and provisioned a private Tailscale mesh VPN to expose internal-only services without public exposure

Open Source Package Maintainer — 2025 - Present
AUR (Arch User Repository) — Remote

- Maintain six packages on the Arch User Repository

Technical Skills

Programming Languages: Python, C++, SQL (Postgres, SQLite, MSSQL), TypeScript, Go
Markup: HTML5, Markdown, Typst, Tailwind CSS
Frameworks: Astro, FastAPI, Hono, React, TanStack Start
Machine Learning: scikit-learn, XGBoost, pandas, NumPy
Infrastructure: Cloudflare, Tailscale, Linux, Caddy
Developer Tools: Git, Docker & Docker Compose, Bash